

Reading the Radar

Glossary and Formula Reference — all five blocks

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Clouds4Africa • July 2026

This is a companion to the five *Reading the Radar* notebooks. Each section collects the specialized terms (with one-line definitions) and the key formulas for one block. Symbols common to several blocks are listed once below. Coefficients are the teaching values used in the widgets; operational relations vary with wavelength, drop-size distribution, and convention.

Common notation

Symbol	Meaning	Symbol	Meaning
c	speed of light (2.998×10^8 m/s)	τ	pulse duration (s)
f	transmit frequency (Hz)	PRF	pulse repetition frequency (Hz)
λ	wavelength, $\lambda = c/f$ (m)	Z	reflectivity factor (mm^6/m^3 , or dBZ)
D	antenna (dish) diameter (m)	R	rain rate (mm/h)
θ	half-power beamwidth (rad or °)	N	drop-number concentration
G	antenna gain	a_e	Earth radius, 6371 km
r	range from radar (m or km)	R_e	effective Earth radius (km)
φ	elevation (tilt) angle	h	beam-centre height (km)

1 Block 1 — The beam in space

Terms

Half-power (3 dB) beamwidth the angular width θ of the beam between the points where transmitted power falls to half its peak; fixed in degrees, so the beam's physical width grows with range.

Antenna gain the concentration of radiated power into the main lobe relative to an isotropic radiator; rises with dish size and falls with wavelength.

Beam height the height of the beam centre above the surface; it rises with range because the Earth curves away faster than the beam bends down.

Standard refraction / 4/3-Earth model the usual downward bending of the beam in a standard atmosphere, modelled by replacing the Earth radius with $\frac{4}{3}a_e$ and treating the beam as straight.

Refractivity N and its gradient dN/dh the scaled refractive index of air; its vertical gradient controls how strongly the beam bends.

Effective Earth radius R_e the inflated Earth radius that lets a curved beam be drawn as a straight line; $R_e = \frac{4}{3}a_e$ under standard refraction.

Anomalous propagation (anaprop) any departure from standard refraction — super-refraction, sub-refraction, or ducting — driven by an unusual refractivity gradient.

Super-/sub-refraction, ducting stronger-than-standard downward bending, weaker bending, and (at $dN/dh < -157$ N/km) bending that traps the beam near the surface, producing ground returns and false echoes.

Beam blockage (PBB, CBB) terrain intercepting part of the beam; *partial* blockage is the fraction blocked at a gate, *cumulative* blockage persists for every gate farther downrange.

Overshooting the lowest beam passing above shallow low-level weather at long range, so it is missed.

Formulas

Beamwidth, width, and gain

$$\lambda = \frac{c}{f}, \quad \theta \approx 1.27 \frac{\lambda}{D} \text{ (rad)}, \quad w = r \theta, \quad G \propto \left(\frac{D}{\lambda} \right)^2.$$

Beam height (4/3-Earth) and its small-angle split

$$R_e = \frac{4}{3}a_e \approx 8495 \text{ km}, \quad h = \sqrt{r^2 + R_e^2 + 2rR_e \sin \varphi} - R_e \approx r \sin \varphi + \frac{r^2}{2R_e}.$$

Refraction, ducting, and blockage bias

$$R_{\text{eff}} = \frac{a_e}{1 + a_e (dN/dh) \times 10^{-6}}, \quad \text{ducting: } \frac{dN}{dh} < -157 \frac{\text{N}}{\text{km}} \iff \frac{dM}{dh} < 0, \quad M = N + 0.157 h.$$

$$\text{reflectivity bias} = 10 \log_{10}(1 - \text{CBB}) \text{ dB}.$$

2 Block 2 — One measurement

Terms

Pulse width (duration) τ the length of the transmitted pulse; it sets range resolution, sensitivity, and blind range together.

Range resolution Δr the minimum down-range separation at which two targets are seen as distinct, $\Delta r = c\tau/2$.

Blind range the near-radar zone the radar cannot listen to while it is transmitting, $\sim c\tau/2$.

Pulse compression transmitting a long coded pulse (for sensitivity) and compressing it on receive (for resolution), breaking the usual trade.

Resolution (sample) volume the box of atmosphere a single measurement averages over: cross-section $r\theta$ by depth $c\tau/2$; its volume grows as r^2 .

Beam filling whether the resolution volume is fully occupied by weather; a partly filled beam reads a value lower than the true peak.

Reflectivity factor Z (dBZ) a measure of how much power weather scatters back, set by the drop sizes and number; expressed logarithmically as dBZ.

Weather radar equation the relation giving received power from transmit power, antenna, wavelength, range, and Z .

Minimum detectable reflectivity (MDS) the weakest Z a radar can detect at a given range; it rises with range.

Distributed vs. point target weather fills the volume (echo falls as r^{-2}); a discrete target does not (echo falls as r^{-4}).

Formulas

Resolution, sensitivity, sample volume

$$\Delta r = \frac{c\tau}{2}, \quad \text{sensitivity} \propto \tau \text{ (+3 dB per doubling)}, \quad V \approx (r\theta)^2 \frac{c\tau}{2} \propto r^2.$$

Weather radar equation and minimum detectable Z (fixed dish)

$$P_r \propto \frac{P_t D^2 \tau Z}{\lambda^4 r^2}, \quad Z_{\min} \propto \frac{\lambda^4 r^2}{P_t D^2 \tau}.$$

Sensitivity rules of thumb

$$\begin{aligned} &+6 \text{ dB per range doubling (distributed targets)}, & \times 2 \text{ power} &= -3 \text{ dB}, \\ &\times 2 \text{ dish} = -6 \text{ dB}, & S \rightarrow X \text{ (fixed dish)} &\approx -21 \text{ dB}. \end{aligned}$$

3 Block 3 — Range, velocity, and ambiguity

Terms

Pulse repetition frequency / time (PRF, PRT) how often pulses are sent ($\text{PRT} = 1/\text{PRF}$); the single knob that sets both range and velocity limits.

Maximum unambiguous range $r_{\max}$ the farthest range whose echo returns before the next pulse; beyond it, echoes are misplaced.

Second-trip echo / range folding an echo from beyond $r_{\max}$ shown at $(\text{true range} - r_{\max})$.

Nyquist velocity v_{nyq} the largest unambiguous radial velocity; faster motion aliases.

Velocity aliasing (folding) a velocity beyond $\pm v_{\text{nyq}}$ wrapping into the window, so fast outbound motion can read as inbound.

Doppler dilemma the fixed trade $r_{\max} v_{\max} = c\lambda/8$: raising PRF buys velocity but spends range, and vice versa.

Dual-PRF / staggered PRT transmitting two interleaved pulse rates and combining them to raise the effective Nyquist beyond the single-PRF curve.

Surveillance vs. Doppler sweep a low-PRF scan for long-range reflectivity (no velocity) and a high-PRF scan for velocity (short range).

Formulas

Range, velocity, and the dilemma

$$r_{\max} = \frac{c}{2 \text{PRF}}, \quad v_{\text{nyq}} = \frac{\lambda \text{PRF}}{4}, \quad r_{\max} v_{\max} = \frac{c\lambda}{8}.$$

$$v_{\text{disp}} = [(v + v_{\text{nyq}}) \bmod 2v_{\text{nyq}}] - v_{\text{nyq}}, \quad v_{\text{ext}} = \frac{v_1 v_2}{|v_1 - v_2|}.$$

4 Block 4 — Wavelength, weather, and the scan

Terms

Attenuation energy absorbed and scattered out of the beam by rain, so weather behind a cell is seen through a curtain; negligible at S, real at C, severe at X.

Specific attenuation k the loss per unit path length, $k = aR^b$ (dB/km), one-way.

Path-integrated attenuation (PIA) the accumulated two-way loss along the beam, $2 \int k dr$.

Wavelength bands (S, C, X) ~ 10 cm, ~ 5 cm, ~ 3 cm; shorter wavelengths are more sensitive but attenuate more.

Detectability crossover the range where a shorter wavelength's sensitivity head-start is overtaken by its accumulated attenuation.

Volume Coverage Pattern (VCP) the stack of elevation tilts a radar scans to build a volume.

Cone of silence the cone above the top tilt that the radar never samples; half-angle $90^\circ - (\text{top tilt})$.

Volume (update) time the time to complete one VCP; grows with the number of tilts.

Phased array an electronically-steered antenna that revisits faster, easing the coverage-vs-time trade.

Z–R relation the empirical link from reflectivity to rain rate used for QPE, e.g. $Z = 200R^{1.6}$.

Formulas

Attenuation, rainfall, and sensitivity gap

$$k = a R^b \text{ (dB/km)}, \quad \text{PIA} = 2 \int_0^r k ds, \quad Z = 200 R^{1.6}, \quad \Delta Z_{\text{min}}^{\text{X-S}} \approx 21 \text{ dB (fixed dish)}.$$

Power with band specifications, and scan timing

$$P_r \propto \frac{P_t}{\theta^2 \lambda^2} \quad (\times 2 \text{ power} = +3 \text{ dB}, \quad \frac{1^\circ}{0.5^\circ} \text{ beam} = +6 \text{ dB}, \quad \lambda^2 = +6 \text{ dB}), \quad t_{\text{vol}} = N_{\text{tilts}} \cdot \frac{60}{\text{rpm}}.$$

Teaching coefficients

Band	λ / freq	specific attenuation a ($k = aR$)	relative sensitivity
S	~ 10.7 cm / 2.8 GHz	$\approx 1 \times 10^{-4}$	reference
C	~ 5.4 cm / 5.6 GHz	$\approx 6 \times 10^{-3}$	more sensitive, attenuates more
X	~ 3.2 cm / 9.4 GHz	$\approx 3 \times 10^{-2}$	$\approx +21$ dB vs. S, attenuates most

5 Block 5 — Two polarizations

Terms

Dual-polarization (polarimetric) radar a radar transmitting and receiving both horizontal (H) and vertical (V) waves; comparing them reveals the shape, variety, and phase of the echo.

Differential reflectivity Z_{DR} the H-to-V power ratio in dB; positive for flattened (large) drops, near zero for round particles (drizzle, tumbling hail). Independent of concentration.

Axis ratio (oblateness) the vertical-to-horizontal diameter ratio of a drop; falls below 1 as drops grow and flatten.

Correlation coefficient ρ_{hv} how alike the H and V returns are across the volume; ≈ 1 for uniform rain, lower for mixtures, very low for non-weather. A purity meter.

Differential phase Φ_{DP} the accumulated H-minus-V phase shift along the path; rises monotonically through rain and is unaffected by attenuation.

Specific differential phase K_{DP} the range slope of Φ_{DP} ; proportional to rain rate and immune to attenuation and calibration.

Differential attenuation the extra loss of the H wave relative to V behind heavy rain, which biases Z_{DR} (but not K_{DP}).

Melting layer (bright band) the layer of melting snow producing intermediate ρ_{hv} and a reflectivity enhancement.

Hydrometeor classification using Z , Z_{DR} , ρ_{hv} , and K_{DP} together to label the echo (rain, big drops, hail, melting layer, non-meteorological).

Formulas

Differential reflectivity (oblate-spheroid model)

$$\gamma \approx 1.03 - 0.062 D \text{ (mm)}, \quad Z_{DR} = 10 \log_{10} \frac{Z_h}{Z_v} = 20 \log_{10} \frac{L_z + k}{L_x + k}, \quad k = \frac{1}{\varepsilon - 1}.$$

Correlation coefficient

$$\rho_{hv} = \frac{|\langle s_h s_v^* \rangle|}{\sqrt{\langle |s_h|^2 \rangle \langle |s_v|^2 \rangle}}, \quad \begin{cases} \rho_{hv} > 0.97 & \text{rain} \\ 0.85\text{--}0.97 & \text{mixed / hail / melting} \\ < 0.85 & \text{non-meteorological} \end{cases}$$

Differential phase and rainfall

$$\Phi_{DP}(r) = 2 \int_0^r K_{DP} ds, \quad K_{DP} = \frac{1}{2} \frac{d\Phi_{DP}}{dr}, \quad R = 44 K_{DP}^{0.82} \iff K_{DP} = \left(\frac{R}{44}\right)^{1/0.82}.$$

References

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Fabry, *Radar Meteorology: Principles and Practice* — intuitive modern treatment.

Bech et al. (2003); Carey et al. (2000) — beam blockage; C-band attenuation correction.

NWS / WDTD training (<https://training.weather.gov>); Py-ART (<https://arm-doe.github.io/pyart>).